

From Retinal Flow to Predictive Visual Fields: Image-Native Language in Continuous State and Pixel Space

Imagized Language Model Project

August 2026

Abstract

Most language models represent language through symbolic tokens and treat visible writing as an auxiliary modality. We study a stricter object: an independent model whose inputs, state supervision, probability model, and primary outputs are continuous images of writing. Earlier Retinal Flow Language Models (RFLM) read ordered 32×32 image fixations, integrate them recurrently, write the next fixation by pixel flow, and reread generated ink; their monolithic writer remains unreadable. We therefore implement a Predictive Visual Field (PVF) that factorizes visual language from visual actuation. A frozen convolutional retina maps writing images to the unit sphere, a causal field integrates image history, a deterministic continuous proposal predicts a low-variance next visual state, and an intrinsic hyperspherical flow models alternatives. V16 adds residual multiscale causal memory with dilated local fields and global causal attention. The 16.47M-parameter student contains no tokenizer, strings after rasterization, Unicode IDs, OCR, character labels, output vocabulary, discrete visual codebook, candidate classifier, or external language model. On a frozen 512-form, four-view Chinese benchmark, selected V16 proposal top-1 is 6.26%, exceeding last-image-only (3.97%), unigram (1.73%), and random dynamics (0.22%); full history adds +0.0773 normalized target log-probability. The stochastic field reaches 3.69%, versus 3.24% last-only, and sampled context cosine gain is +0.0795. The model trains on one RTX 4090 with 1.479 GiB peak allocated CUDA memory. These results establish causal visual-state learning under a strict image-only boundary, but remain below a symbolic bigram (13.14%). We then train a separate 5.73M-parameter V17 visual actuator from continuous retinal state and style image to ink pixels. On one untouched 512-example evaluation, correct-state visual identity is 58.59% versus 0.98% after shuffling only intended states; target cosine is 0.713 versus 0.086. However, pixel F1 is 0.438, below the fixed 0.500 gate, and human review rejects readability. Thus a small continuous visual state causally controls generated writing on one consumer GPU. A topology-first V18 planner then maps the same kind of continuous intent and a style image to a directly supervised spatial ink plan. Its 2.36M trainable parameters reach 73.63% identity top-1 versus 0.98% shuffled and pixel F1 0.658 versus 0.313 on a fresh 512-example development audit, while using 0.778 GiB peak allocated CUDA memory. Most reviewed simple and medium forms are recognizable, but dense forms remain malformed. The V18 frozen bank remains sealed because the human gate lacked a prespecified numeric rubric. These experiments refute a categorical impossibility claim about compact continuous-image writing, while autonomous language, broad readability, historical question answering, and efficiency over text models remain unproved.

1 Motivation

Human readers do not see byte-pair tokens. They see pages, lines, strokes, spacing, damage, font variation, and script-specific visual forms. Current language models are extremely capable, but

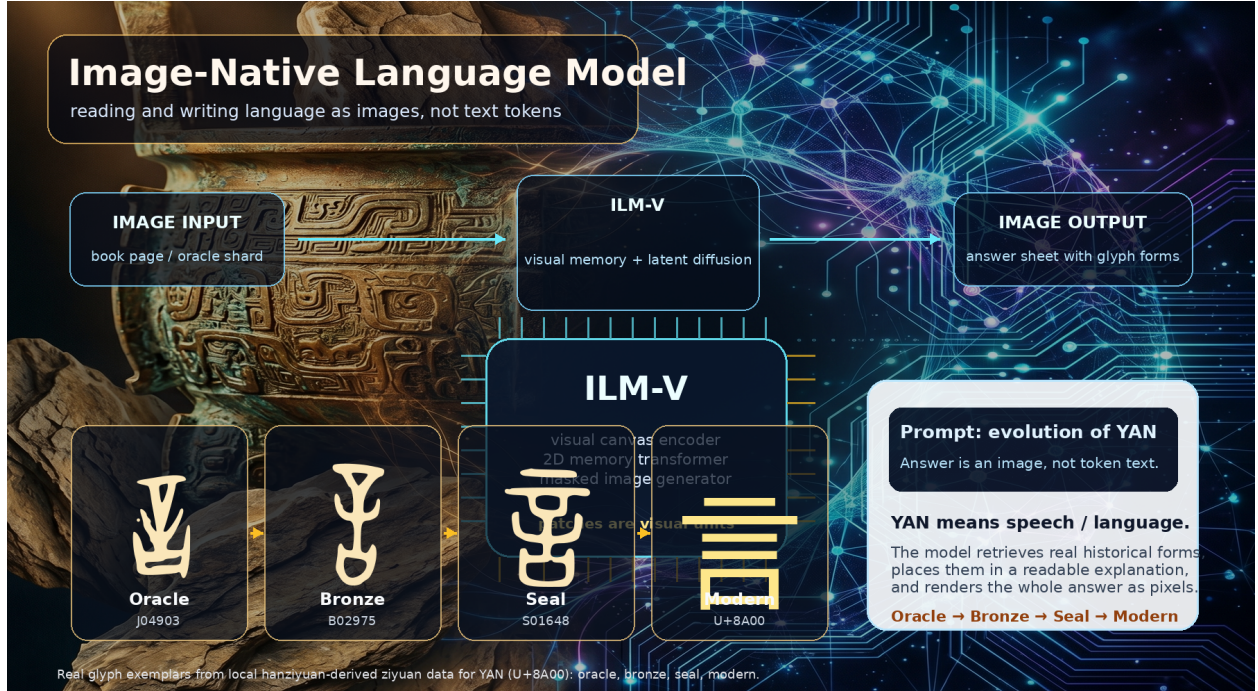


Figure 1: Image-native language modeling concept figure generated with an AgInTi background and overlaid with real local ziyuan glyph exemplars for YAN (U+8A00). The answer is a rendered image containing both modern explanation and historical forms.

their central representation is normally a sequence of text tokens. Multimodal models accept images, and modern APIs also expose image-generation endpoints. For example, OpenAI documents APIs for processing images as input and generating images as output, and Google’s Gemini API documentation describes image generation from text and image prompts [4, 5, 6]. These systems are important baselines, but they do not by themselves define a language model whose primary language substrate is the visible page.

The Imagized Language Model (ILM) goal is different:

$$\text{image input} \rightarrow \text{visual language state} \rightarrow \text{image output}.$$

The output should itself be a readable answer image. For a query such as “explain the evolution of character yan”, the system should produce a page containing modern explanatory writing, classical-style Chinese, and oracle, bronze, seal, and modern forms. Typed text can still be accepted by a user interface, but it is rasterized into an image prompt before entering the core model. Conversely, the model’s answer is a page image first; OCR or transcription is an optional downstream extraction step.

2 Prior Art and Gap

OCR-free document models such as Donut and screenshot parsing models such as Pix2Struct show that document images can be understood without an explicit OCR pipeline. PIXEL reconstructs masked rendered-text patches and demonstrates cross-script visual transfer [10]; PIXAR is a genuinely generative pixel-space autoregressive language model and identifies noisy maximum-likelihood text images as a central difficulty [11]; PixelGPT studies autoregressive visual and textual

Retinal Flow Language Model

Read visible fixations, predict a visual distribution, write ink, then read the model's own ink.

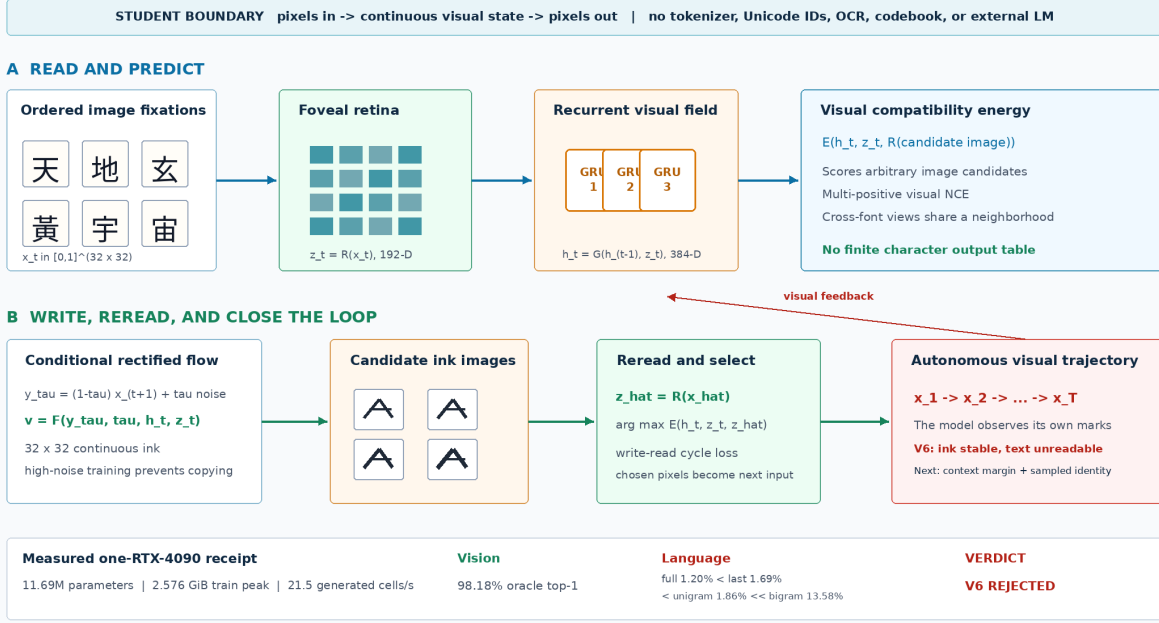


Figure 2: The implemented Retinal Flow Language Model. Training joins visual next-state energy, conditional rectified flow, a write-read cycle, V6 model-induced visual rollouts, and V7 calibrated context and sampled-image identity. Autonomous inference samples candidate ink images, rereads them, selects by visual energy, and feeds the selected pixels back.

pretraining [12]. These are the closest model-level precedents. Token-free language models such as CANINE, ByT5, and MEGABYTE remove subwords but retain symbolic bytes or characters. Discrete diffusion language models show that generation need not be left-to-right, but usually retain a linguistic vocabulary.

I-JEPA shows that predicting target image representations from visual context can learn semantic visual fields without reconstructing every pixel [26]. Flow matching trains continuous generative paths by vector-field regression and supports efficient numerical sampling [23]. Human reading is also sequential and foveated: Chinese readers make saccades and can use both character- and word-related targeting cues [33]. These observations motivate RFLM, but they do not imply biological equivalence.

The remaining gap is not simply “pixels instead of tokens.” It is a system whose predictive random variables are continuous writing images, whose memory is learned from ordered visual observations, whose writer produces new pixels, and whose own output distribution is present during evaluation. Historical evidence adds a separate constraint: synthesized marks must never be presented as attested source glyphs.

3 Architecture

Figure 2 gives the strict visual contract. For fixation images $x_t \in [0, 1]^{H \times W}$, RFLM factors a visual sequence as

$$p_{\Theta}(x_{1:T}) = p(x_1) \prod_{t=1}^{T-1} p_{\Theta}(x_{t+1} \mid x_{\leq t}).$$

The random variables are continuous image fields rather than entries in a linguistic or visual vocabulary.

Foveal retina. A convolutional reader maps each 32×32 image to a normalized 192-dimensional visual state:

$$z_t = R_{\theta}(x_t) / \|R_{\theta}(x_t)\|_2.$$

Independent font renderings are aligned with visual contrastive, invariance, and variance objectives. The target retina $R_{\bar{\theta}}$ is an exponential moving average of the online retina. Font coverage is checked from font cmap tables so missing-glyph boxes cannot become a spurious class.

Recurrent visual field and energy. A three-layer GRU integrates the fixation history,

$$h_t = G_{\phi}(h_{t-1}, z_t), \quad h_t \in \mathbb{R}^{384}.$$

An energy function scores any candidate image c_j after rereading it:

$$s_{t,j} = E_{\psi}(h_t, z_t, R_{\bar{\theta}}(c_j)).$$

Multi-positive visual NCE admits multiple near-identical image views as valid targets. Unlike a softmax language head, this energy is not tied to a finite character inventory.

Conditional ink flow. For clean target x_{t+1} , noise ϵ , and $\tau \in [0, 1]$, define

$$y_{\tau} = (1 - \tau)x_{t+1} + \tau\epsilon, \quad u_{\tau} = \epsilon - x_{t+1}.$$

The pixel writer predicts the path velocity

$$\hat{u}_{\omega} = F_{\omega}(y_{\tau}, \tau, h_t, z_t), \quad \mathcal{L}_{\text{flow}} = \|\hat{u}_{\omega} - u_{\tau}\|_2^2.$$

Training emphasizes high-noise times, where target pixels cannot simply be copied, and weights sparse strokes and the predicted clean endpoint.

Write-read cycle. The one-step endpoint $\hat{x}_0 = y_{\tau} - \tau\hat{u}_{\omega}$ is reread through the frozen target retina. A visual NCE cycle objective aligns it to independent target renderings:

$$\mathcal{L}_{\text{cycle}} = \text{NCE}\left(R_{\bar{\theta}}(\hat{x}_0), R_{\bar{\theta}}(x_{t+1}^{(1)}), R_{\bar{\theta}}(x_{t+1}^{(2)})\right).$$

The complete loss combines flow, energy, cross-render invariance, retina contrast, endpoint, stroke, and cycle terms. At inference, the writer samples a candidate set $\mathcal{C}_t$ and selects

$$\hat{x}_{t+1} = \arg \max_{c \in \mathcal{C}_t} E_{\psi}(h_t, z_t, R_{\bar{\theta}}(c)).$$

The selected pixels become the next observation. This closes the visual loop and turns autonomous stability into a mandatory model property rather than a post-processing concern.

Model-induced visual rollout. V6 invokes the same sampler and energy reranker during training. From a clean prefix it generates two short-flow candidates, selects and detaches one bitmap, rereads it with the online retina, and advances the recurrent state. For two rollout steps it minimizes

$$\mathcal{L}_{\text{roll}} = 0.15\mathcal{L}_{\text{state}} + 0.35\mathcal{L}_{\text{energy}}^{\text{roll}} + 0.30\mathcal{L}_{\text{recovery}}.$$

The terms align clean and induced recurrent states, predict the real next image from the induced state, and recover clean next pixels after generated feedback. All targets remain continuous image fields. Candidate selection is stop-gradient, which bounds memory while exposing the trainable state and writer to deployed pixels.

Calibrated context and sampled identity. V7 corrects two training/evaluation mismatches. Raw target-energy differences can be increased by changing the scale or all competing scores, so V7 uses the normalized visual target log probability

$$\ell(y \mid h, \mathcal{C}) = s(y \mid h) - \log \sum_{c \in \mathcal{C}} \exp s(c \mid h)$$

and minimizes

$$\mathcal{L}_{\text{ctx}} = \max\{0, m - \ell(y \mid h_{\text{full}}, \mathcal{C}) + \ell(y \mid h_{\text{last}}, \mathcal{C})\}.$$

The last-only branch is detached. The training-only set $\mathcal{C}$ contains independent image renderings; positives are inferred by retinal similarity. Offline strings select the curriculum but anchor IDs and target indices do not enter the student, the pixels are disjoint from evaluation views, and the anchors are not deployed. V7 also differentiates through a two-step numerical pixel-flow endpoint,

$$\mathcal{L}_{\text{sample}} = \text{NCE} \left(R_{\bar{\theta}}(\hat{x}_0^{K=2}), \{R_{\bar{\theta}}(y^{(v)})\}_{v=1}^V \right).$$

This trains the sampled image path rather than only a one-step endpoint proxy.

4 Predictive Visual Field

Figure 3 separates linguistic uncertainty from stroke rendering. For target state $z_{t+1} = R(x_{t+1}) \in \mathbb{S}^{d-1}$, a deterministic proposal predicts

$$\mu_t = \frac{P_\psi([h_t, z_t])}{\|P_\psi([h_t, z_t])\|_2}.$$

The proposal is a continuous image-derived state, not a class index. It is trained with geodesic alignment, image contrast, and full-history advantage against a detached last-image branch.

In parallel, choose random unit source ϵ , let q_τ follow the shortest spherical geodesic from z_{t+1} to ϵ , and let u_τ be its tangent velocity. The conditional field learns

$$\mathcal{L}_{\text{state-flow}} = \mathbb{E} \|v_\eta(q_\tau, \tau, [h_t, z_t]) - u_\tau\|_2^2.$$

The loss is full tangent-vector energy, not coordinate-mean MSE. Sampling integrates backward with the spherical exponential map,

$$q_{k-1} = \text{Exp}_{q_k}(-\Delta\tau v_\eta(q_k, \tau_k, [h_t, z_t])),$$

Predictive Visual Field

Language evolves as a distribution over continuous retinal states; a separate visual actuator writes each state as ink.

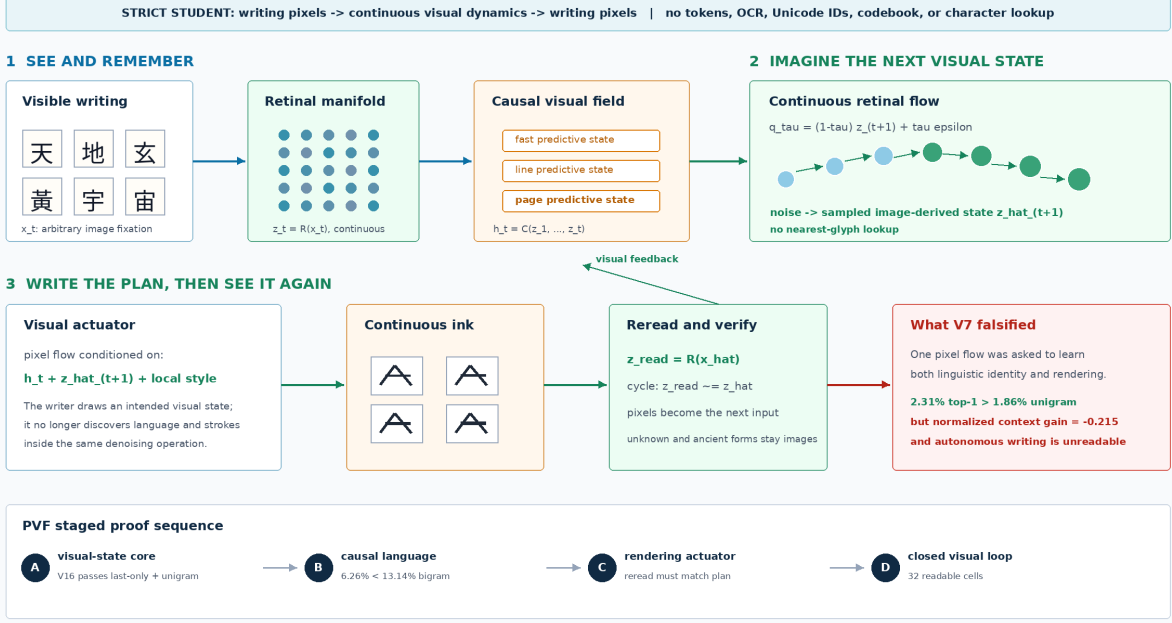


Figure 3: Predictive Visual Field factorization. A causal image field predicts a low-variance continuous visual proposal and a stochastic distribution over image-derived retinal states. A later separate actuator renders a selected state and the retina rereads its pixels.

yielding intended next visual states $\hat{z}_{t+1}$, not glyph IDs. Candidate images are used only by the evaluator: it rereads them with the frozen retina and computes kernel density against continuous samples. There is no learned 512-way output layer.

The training-only multiview anchor curriculum contains image tensors. Positives are inferred from retinal cosine similarity; labels and target indices do not enter the student, and the bank is not deployed. This visual contrast is necessary because geodesic endpoint alignment alone can reward a visual mean without separating language alternatives.

V16 augments the pretrained recurrent trajectory $h_{1:t}^R$ with three residual causal memory blocks. Each block combines a left-padded depthwise temporal field at dilation 1, 2, or 4, global causal attention, and a gated feed-forward residual. For memory M_ϕ , the corrected state is

$$h_t^{V16} = h_t^R + \sigma(g)W_o \text{LN}\{M_\phi(h_{1:t}^R, z_{1:t})\}.$$

The per-channel gate is initialized at $g = -4$, preserving the trained V15 function at the start of the upgrade. The selected checkpoint has mean $\sigma(g) = 0.018023$; therefore the frozen result is a bounded memory pilot, not evidence that a large residual correction is necessary.

V17 implements a separate visual actuator that writes

$$\hat{x}_{t+1} = W_\omega(\epsilon_x, h_t, \hat{z}_{t+1}, R_\theta(x_t))$$

and is trained with pixel flow plus a reread constraint

$$\mathcal{L}_{\text{act}} = \mathcal{L}_{\text{pixel-flow}} + \lambda [1 - \cos\{R_{\hat{\theta}}(\hat{x}_{t+1}), z_{t+1}\}].$$

PVF V16: Causal Visual Memory on One RTX 4090

A 16.47M-parameter student reads writing pixels, forms continuous visual states, and predicts the next state without a symbolic vocabulary.

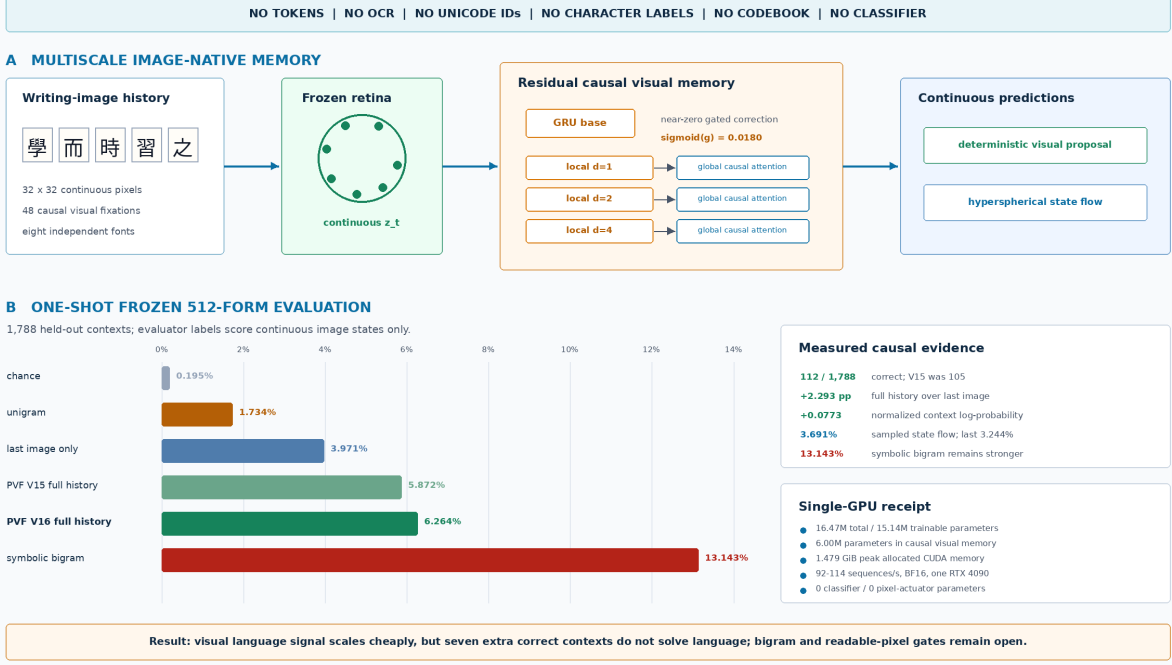


Figure 4: Measured PVF V16 memory result. Both deterministic proposal and stochastic hyperspherical field use full image history and beat random, last-only, and unigram controls. V16 improves proposal top-1 by seven frozen contexts over V15, but remains below the symbolic bigram and has no pixel actuator.

No nearest-glyph projection, character classifier, or token unembedding is permitted. In V17, $\hat{z}_{t+1}$ is supplied by a different-font image of the target form so actuation is isolated from language prediction. This differs from Embedded Language Flows, which demonstrates effective continuous flow dynamics but encodes and finally unembeds discrete tokens [29]. It combines the next-fixation representation objective of recurrent JEPA [27] with the prediction/generation separation explored by D-JEPA [28], and uses intrinsic flow matching on a Riemannian manifold [24, 25], while retaining an image-derived language state.

5 Isolated Visual Actuation and Motor Planning

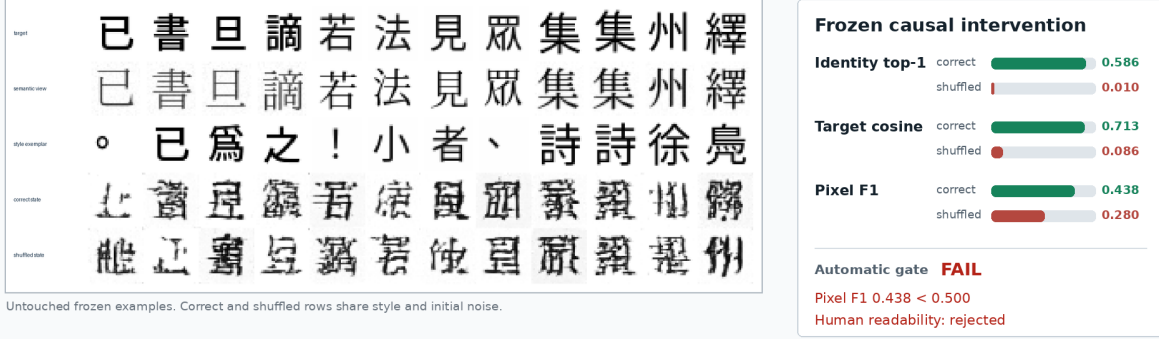
The V17 actuator tests whether an image-derived continuous state can control new pixels without a glyph table. A frozen retina produces $z = R(x^{\text{semantic}}) \in \mathbb{S}^{191}$ from one font view, and a convolutional encoder produces style vector $s = E_{\omega}(x^{\text{style}})$ from a different character in the desired target render. A 5.73M-parameter conditioned U-Net integrates rectified flow from Gaussian noise to a 32×32 ink image. The target image enters losses only; the spatial condition to the writer is a constant blank plane.

Training combines stroke-weighted flow matching, an endpoint loss, retinal cosine, multi-positive retinal contrast, and gradients through a two-step deployed sampler. At evaluation, correct and shuffled branches receive identical style images and initial noise. Only the intended state is rolled

V17: continuous visual state controls pixels, not yet readable strokes

One RTX 4090 | 5.73M trainable actuator parameters | no token IDs, OCR, Unicode IDs, labels, or glyph lookup

different-font image -> frozen retina -> continuous intended state + style image -> pixel flow -> generated ink -> reread



Measured verdict visual identity is causally controlled; exact stroke topology and readability are not solved

Next: decode the continuous state into a supervised spatial motor plan, then use stochastic flow only for refinement

Figure 5: Measured V17 frozen result. Continuous intended state strongly controls recovered visual identity under a same-noise shuffled-state intervention, but pixel F1 and human readability fail. The correct-state row contains state-dependent pseudo-characters, not reliably readable target forms.

within the batch. The global evaluator retrieves among all 512 generated examples, with duplicate positives inferred from image-state similarity rather than labels.

V17 was selected at step 1,600 by a preregistered development rule and queried the frozen record split once. Correct-state top-1 is 58.594%, versus 0.977% shuffled; target cosine is 0.71301 versus 0.08612; and style-copy cosine is only 0.05858. These measurements reject style copying as the explanation for the causal signal. Pixel F1 is 0.43849 versus 0.28001 shuffled, failing the required 0.50, and human review rejects most outputs as unreadable. Training takes 339.67 seconds and peaks at 1.588 GiB allocated CUDA memory on one RTX 4090.

The result identifies a topology gap rather than an identity-control gap.

5.1 Topology-first visual motor plan V18

V18 decodes state and style into a directly supervised spatial visual motor plan,

$$p_\omega = D_\omega(z, E_s(x^{\text{style}})) \in [0, 1]^{H \times W}.$$

A context MLP modulates a learned 4×4 seed, conditioned residual blocks upsample it to 32×32 , and a sigmoid emits continuous ink. The loss joins stroke-weighted binary cross entropy, soft Dice, pixel and Sobel-edge distances, retinal identity and contrast, and a correct-versus-shuffled state margin. Target pixels supervise the output but never enter the condition. The plan remains continuous and image-derived; it is not copied target ink, an ID lookup, or a discrete visual codebook. Stochastic flow is reserved for later residual style and surface variation rather than stroke placement.

V18: a 2.36M visual motor plan learns readable writing topology

One RTX 4090 | continuous image-derived intent | no tokens, OCR, Unicode IDs, labels, lookup, or codebook

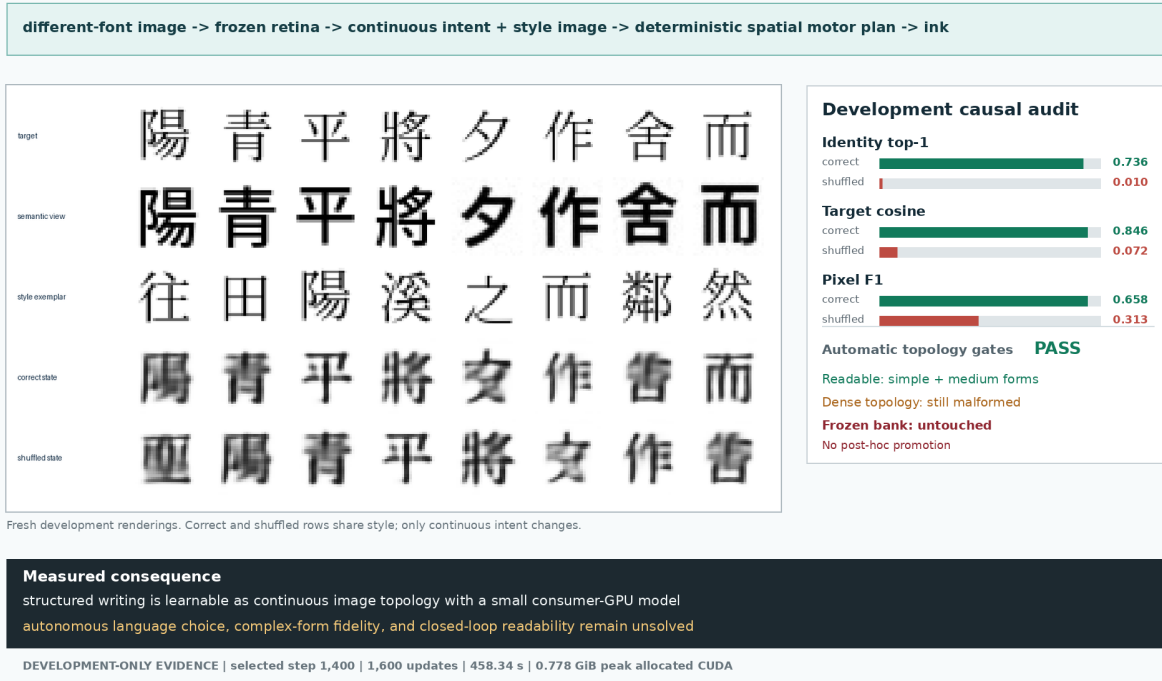


Figure 6: Measured V18 development result. A 2.36M-parameter deterministic motor planner turns continuous image-derived intent into recognizable writing. Shuffling only intended state destroys identity and topology while preserving mean ink occupancy. This is development-only evidence; the frozen bank remains untouched.

The salted partition contains 6,573 training, 223 development, and 221 frozen records. Selection maximized development correct-state pixel F1 subject to fixed topology, state-dependence, retinal-similarity, and non-copying gates. Step 1,400 was selected at pixel F1 0.6890 versus 0.3368 shuffled and global identity top-1 78.32% versus 1.17%. A fresh development seed yields 512 examples: identity top-1 is 73.63% versus 0.98%, target cosine is 0.8462 versus 0.0716, and pixel F1 is 0.6577 versus 0.3129. Mean ink fraction is 0.17505 in both branches and style-copy cosine is 0.0686.

Full-resolution review finds clear simple and medium forms but merged strokes in dense forms. Because the preregistration required human readability without defining a numeric blinded threshold, post-hoc inspection cannot authorize a frozen query. V18 therefore accepts readable topology emergence on development, rejects broad-complexity readability, and leaves the frozen identifier bank sealed. The next experiment must define complexity strata and a recognition rubric prospectively.

6 Data

The project already has a strong historical Chinese glyph source outside the tracked repo. A local snapshot at `/home/lachlan/ProjectsLFS/incoder/data/historic` contains:

Item	Count
Characters	9,055
Glyph records	84,642
Oracle	32,216
Bronze	23,500
Liushutong	21,923
Seal	6,996

This supports future evidence-bearing historical answer pages but was not used as a character-label channel in the RFLM run. The language trajectory corpus is a 7,017-record manifest built from 16 public-domain Chinese Wikisource book packages. The contribution layer is recorded as CC BY-SA, and the manifest stores source artifacts and rights. Strings exist only in the offline manifest and renderer; model batches contain image tensors.

Eight Noto Sans/Serif CJK fonts provide independent visual views. The run stores the exact font paths, byte counts, and SHA-256 hashes in checkpoints and receipts. Cmap intersection filtering leaves 8,282 supported visible characters; 107 unsupported unique forms account for only 0.0074% of visible corpus instances and are skipped instead of rendered as missing-glyph boxes.

7 Training Objectives

Each 48-fixation page supplies eight sampled causal positions, making the language objective denser than one endpoint per page. The complete training loss is

$$\mathcal{L} = \lambda_f \mathcal{L}_{\text{flow}} + \lambda_e \mathcal{L}_{\text{energy}} + \lambda_r \mathcal{L}_{\text{retina}} + \lambda_v \mathcal{L}_{\text{view}} + \lambda_p \mathcal{L}_{\text{endpoint}} + \lambda_s \mathcal{L}_{\text{stroke}} + \lambda_c \mathcal{L}_{\text{cycle}} + \rho(t) \mathcal{L}_{\text{roll}}.$$

After step 5,200, V7 adds a ramped correction

$$\mathcal{L}_{V7} = \mathcal{L} + \kappa(t) (0.5 \mathcal{L}_{\text{ctx}} + 0.2 \mathcal{L}_{\text{sample}}).$$

The energy term uses 512 cross-render candidate views per batch in the final run. Visual duplicate positives are discovered by image similarity rather than labels. The view term aligns independent renderings in both retinal and candidate-projection spaces. Endpoint and stroke terms improve sparse-ink geometry. The cycle term is weighted toward high-noise examples, where a writer must use recurrent context rather than visible target pixels.

For V6, $\rho(t)$ ramps from zero to one over 400 updates. A rollout subset of eight sequences uses two generated steps, two candidates, and two flow steps per candidate. This follows the distribution-shift lesson of scheduled sampling and DAgger [31, 30]; it remains image-only because the visited states, selected observations, and recovery targets are pixels and continuous visual fields. The final rollout state cosine reaches 0.961, but selected-image target cosine is only 0.103. This distinction predicts the measured outcome: state recovery becomes robust while sampled visual identity remains weak.

V7 resumes V6 for 800 updates with a 512-form, four-view image-anchor curriculum and a two-step differentiable endpoint on four examples per batch. The anchor retina is refreshed every 200 updates. The selected step 5,800 checkpoint is chosen from preregistered validation checkpoints because it has the strongest combination of independent-anchor context gain (+0.1798) and sampled pixel F1 (0.3636). The continuation adds approximately 261 seconds of training and peaks near 3.0 GiB allocated VRAM on one RTX 4090.

7.1 PVF state objectives

V8–V16 reuse the proven retina but remove the pixel writer from the state ablation. The causal GRU, proposal, and hyperspherical field are trainable. For proposal μ , target image state z^+ , and image-derived candidate states $\{z_j\}$, visual identity uses

$$\mathcal{L}_{\text{proposal-NCE}} = -\log \frac{\sum_{j \in \mathcal{P}} \exp\{s\mu^\top z_j\}}{\sum_j \exp\{s\mu^\top z_j\}},$$

where positives $\mathcal{P}$ are inferred by frozen retinal similarity. A detached last-image condition supplies the calibrated context objective

$$\mathcal{L}_{\text{proposal-ctx}} = \max\{0, m - \ell(z^+ | h_{\text{full}}) + \ell(z^+ | h_{\text{last}})\}.$$

Equivalent identity and context losses are applied to states numerically sampled from the flow. Direct sampled geodesic loss prevents inference-path drift, while contrastive loss prevents collapse to the retinal mean.

V15 has 10,470,273 total parameters, 9,137,345 trainable parameters, and zero classifier or pixel-actuator parameters. It trains in BF16 with 48-image sequences and peaks at 1.181 GiB allocated CUDA memory. The full V14–V15 trajectory records 1,895.24 seconds on one RTX 4090. Step 2,000 was selected on the development image bank before running the frozen evaluator.

V16 resumes selected V15 step 2,000, adds 6,001,536 causal-memory parameters, and resets the optimizer with a new 100-step warmup. It has 16,471,809 total parameters, 15,138,881 trainable parameters, and zero classifier or actuator parameters. The 1,200-update continuation records 2,047.63 seconds, about 92–114 sequences per second, and 1.479 GiB peak allocated CUDA memory. Step 2,200 was selected before the frozen evaluator by maximum development proposal top-1 subject to full-context over last-only, positive normalized context gain, and target cosine above 0.10.

8 Image-First Training and Inference

Figure 7 makes the training contract explicit. Typed corpora are rendered into page images; scanned books and manuscripts are cropped and tiled; historical glyph files become glyph tiles; and marks that are missing from font tables remain valid image patches. Metadata such as crop boxes and stage labels may remain in manifests, but the model path should learn from visual latents. OCR is useful as a readability critic, not as the hidden language channel.

Figure 8 shows the user-facing path. A typed prompt is converted into a visual prompt canvas, and an uploaded image is normalized into the same visual space. The model returns a page-like image containing modern language and historical glyph regions. A later OCR/transcoder can attach a searchable text layer for English, Chinese, or Unicode-representable spans; ancient or unencoded glyphs remain image regions with coordinates and provenance.

The formal contract is:

$$\begin{aligned} r_\alpha(\text{text prompt}) &\rightarrow x_{\text{prompt}}, \\ g_\beta(\text{uploaded page or glyph}) &\rightarrow x_{\text{prompt}}, \\ f_\theta(x_{\text{prompt}}, x_{\text{context}}) &\rightarrow y_{\text{page}}, \\ o_\psi(y_{\text{page}}) &\rightarrow \text{optional text layer} + \text{glyph-region metadata}. \end{aligned}$$

Here r_α is only a UI rasterizer, not the language model itself. The core model f_θ maps visual inputs to a rendered page image y_{page} . The optional reader o_ψ may recover Unicode text where possible, but unencoded historical forms remain valid image regions.

A useful answer canvas should resemble a page excerpt from a book or corpus:

Training ILM-V: Language as Visual Corpus

Every source is converted into page/glyph images. The model learns to predict and generate visual latents, not BPE or word tokens.

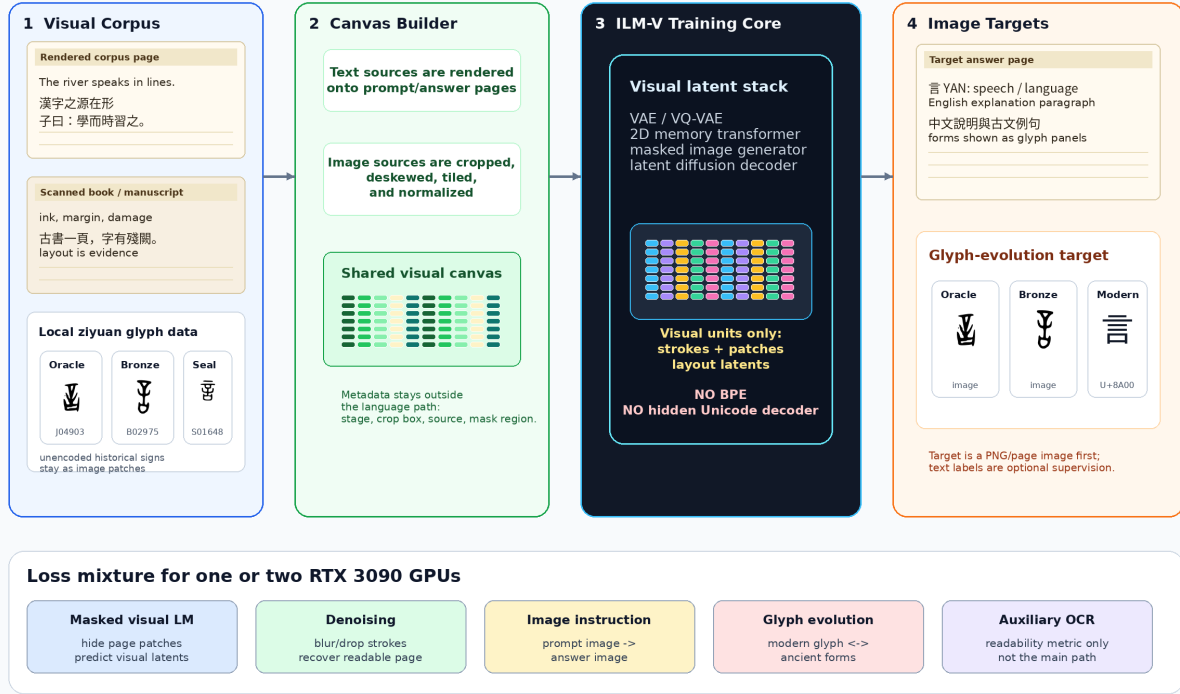


Figure 7: Training pipeline for ILM-V. Rendered text corpora, scanned pages, historical glyph files, and unencoded marks are all converted into visual canvases. The training target is visual latent prediction and image generation, not BPE-token prediction.

- a title region rendered as text image,
- one or more modern-language paragraphs, for example English and Chinese,
- a classical-style line or quotation when appropriate,
- glyph panels for forms that may not exist in computer fonts,
- a provenance strip with source IDs or nearest retrieved exemplars.

This schema is intentionally different from “text answer plus illustrations”. The generated page image is the primary output object, and the text layer is secondary.

9 3090-Friendly Curriculum

Figure 9 records the original page-model schedule. Measurements now support a different order:

1. Learn a cross-font foveal visual alphabet and audit missing-glyph coverage.
2. Train recurrent visual energy and conditional ink flow jointly on dense causal positions.
3. Reread generated endpoints during training and evaluate on a fixed external visual bank.

Inference ILM-V: Chat-Like UI, Image-First Output

Typed text and uploaded images are both turned into visual prompt canvases. The model returns a rendered page image first.

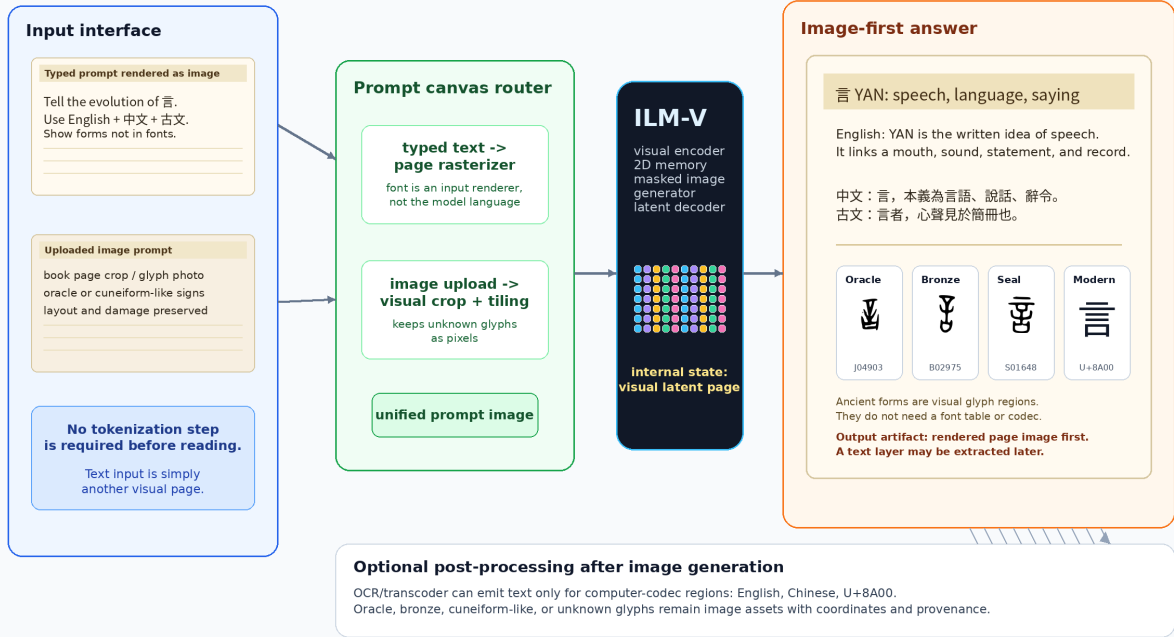


Figure 8: Inference pipeline for ILM-V. A chat-like interface may accept typed text or uploaded images, but typed text is first rasterized into a prompt image. The primary output is a rendered page image. OCR or transcoding is optional and only applies to output regions that exist in computer codecs.

4. Train on autonomous model-induced visual trajectories; V6 completes this step and stabilizes ink but remains unreadable.
5. Train normalized context advantage on independent images and identity through differentiable sampled endpoints; V7 improves both signals but remains unreadable.
6. Train continuous proposal and next-retina flow alone; V15 passes random, last-only, unigram, context-use, and target-signal state gates.
7. Add residual multiscale causal visual memory; V16 replicates the gates with a seven-context proposal increase but does not beat the symbolic bigram.
8. Run a paired memory ablation on a newly preregistered bank before attributing the small V16 change to the memory module.
9. Condition a separate actuator on continuous visual plans; V17 passes isolated causal reread identity but fails topology and readability.
10. Decode a directly supervised spatial motor plan before stochastic refinement; V18 passes automatic development topology gates and makes many forms readable, while dense forms still fail.

3090-Friendly Training Curriculum

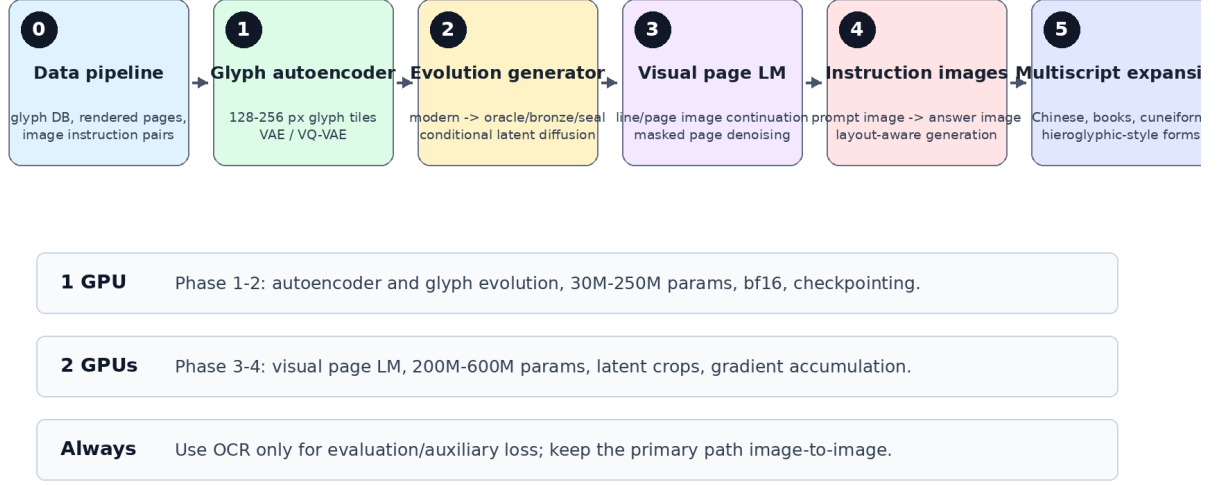


Figure 9: A staged curriculum that fits one or two RTX 3090 GPUs by starting with glyph tiles and latent generation before scaling to page-level image language modeling.

11. Preserve a continuous spatial retinal field, preregister complexity strata and a blinded readability threshold, and open a new frozen bank only after those development gates pass.
12. Add provenance-gated historical panels and instruction images only after readable autonomous continuation.
13. Scale width and corpus size only after the fixed language and stability gates pass.

The 16.47M PVF V16 state model, 5.73M V17 actuator, and 2.36M V18 motor planner are deliberately small enough for one RTX 4090. Their measured peaks are 1.479, 1.588, and 0.778 GiB, respectively, leaving substantial capacity headroom for controlled corrections. V18 makes many isolated forms readable, but this is not an end-to-end efficiency victory: the causal language core remains below a symbolic bigram, dense topology remains imperfect, and autonomous continuation has not passed. “More efficient than an LLM” remains an unsupported hypothesis until quality is matched on a named task.

10 Target Demonstration

The first demo should answer a rendered visual prompt such as: “Explain the evolution of the Chinese character yan.” The target answer image should contain:

- a title and concise modern explanation,
- English and Chinese paragraphs rendered as page text,

Example Output: Evolution of Chinese Character Zhong

A target answer is itself an image: explanation plus historical forms.



Figure uses local hanziyuan-derived glyph files when available; generation targets should retrieve/cite real exemplars.

Figure 10: Target output style for a glyph-evolution question. The answer is not merely text: it is a readable image with historical visual forms. The example uses local hanziyuan-derived glyph files when available.

- a classical-style line such as “yan is the visible trace of speech”,
- a historical timeline with real oracle, bronze, seal, and modern glyph exemplars,
- stage labels and citations as rendered image text.

The existing Zhong figure in Figure 10 is a compact stage-timeline example; the Yan figures in Figures 1 and 8 show the fuller answer-page style. Together they test whether the model can combine language understanding, visual page composition, and historical script generation.

11 AgInTi Artifact Loop

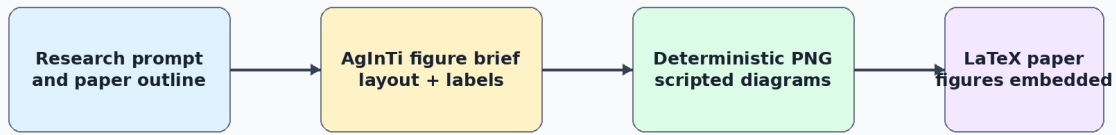
The figure brief for AgInTi-style refinement is stored in the local AgInTi prompts subfolder. The current paper uses scripted PNG generation to keep the artifact reproducible.

12 Evaluation

The model should be evaluated as both a language model and a visual generator:

- Fixed-bank top-1, top-5, and reciprocal rank for full context and a last-fixation ablation.
- Random-weight, chance, unigram, and symbolic bigram baselines on identical contexts.
- Cross-font retina-oracle retrieval, separated from recurrent language prediction.
- Generated-image target similarity, pixel F1, ink occupancy, variance, and energy-reranked hit rate.
- Autonomous rollout length, throughput, VRAM, and blinded human readability.

AgInTi-Assisted Research Artifact Loop



Why scripted PNGs now?

They are reproducible, editable, and compile cleanly in the paper while leaving room for later AgInTi/image-model refinement

Figure 11: AgInTi-assisted artifact loop. The current figures are deterministic scripted PNGs so the paper is reproducible; the figure brief can later be sent to an interactive AgInTi/image agent for visual refinement.

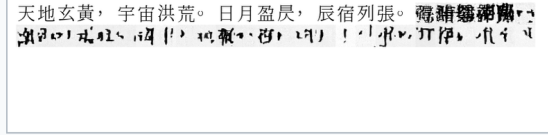
- Student-boundary audit for token IDs, Unicode IDs, OCR, codebooks, labels, and external models.
- Later historical tasks: glyph-stage accuracy, attested-source provenance, and explicit synthetic-form labeling.

Closed-loop visual trajectory training: what changed

Matched 32-cell autonomous generation; same prompt, seed, sampler, and 11.69M-parameter model

V5 Clean-prefix training

Ink thins under its own visual feedback



Rendered prompt followed by 32 model-generated image cells

Late / early ink

0.483

Sparse cells

37.5%

Frozen-bank top-1

0.908%

Generated context signal

+0.0211

V6 Model-induced rollout training

Ink survives, but the writing remains unreadable



Rendered prompt followed by 32 model-generated image cells

Late / early ink

1.168

Sparse cells

18.8%

Frozen-bank top-1

1.197%

Generated context signal

+0.0077

Measured verdict closed-loop stability improved; contextual language and readable generation did not

Frozen glyph bank SHA-256 543987ddd754135cff34dbec0f6f91ce1971741d0807754c65650031121181c4

Figure 12: Matched autonomous evidence for V5 and V6. The prompt, seed, sampler, model size, and 32-cell horizon are fixed. Model-induced rollout training prevents the earlier loss of ink and halves the sparse-cell fraction, but neither continuation is readable and generated target signal weakens.

System	Measured signal	Interpretation
Whole-page flow	velocity MSE 0.851	Unreadable page texture; negative baseline
Global visual memory	top-1 2.23% over 2,016	Above random, but 0/16 historical paraphrases
Causal InkStream	validation ink F1 0.639	Autonomous repeated-grey collapse; negative baseline
RFLM V5 retina oracle	top-1 97.65%	Strong cross-font visual alphabet
RFLM V5 recurrent state	top-1 0.91%	Below last-only 1.36%, unigram 1.86%, and bigram 13.58%
RFLM V6 recurrent state	top-1 1.20%	Improved, but below last-only 1.69% and both language baselines
RFLM V6 pixel writer	context gain +0.0077	Positive but weaker than V5's +0.0211
RFLM V6 autonomous loop	ink ratio 1.168	Stable character-scale marks remain unreadable
RFLM V7 recurrent state	top-1 2.31%	Above last-only 2.02% and unigram 1.86%, far below bigram 13.58%
RFLM V7 calibration	log-probability gain -0.2155	Much better than V6's -0.9066, but full history still lowers mean target probability
RFLM V7 pixel writer	context cosine +0.0303	Passes generated-signal gate; autonomous writing remains unreadable

PVF branch	Frozen signal	Interpretation
V14 proposal	top-1 2.52%	First proposal gate pass; above last-only 2.24% and unigram 1.73%
V14 state flow	top-1 2.35%	First stochastic state gate pass
V15 proposal	top-1 5.87%	Above last-only 4.42%, unigram 1.73%, and random 0.17%; context gain +0.0707
V15 state flow	top-1 3.41%	Above last-only 2.68% and unigram; sampled context cosine gain +0.0805
V16 proposal	top-1 6.26%	112/1,788; above last-only 3.97% and V15 by seven contexts; context gain +0.0773
V16 state flow	top-1 3.69%	Above last-only 3.24% and unigram; sampled context cosine gain +0.0795

Table 1: Frozen Predictive Visual Field results. V15 and V16 branches pass their visual-state gates but remain below the 13.14% symbolic bigram.

V17 frozen actuator metric	Correct state	Shuffled state	Gate
Global identity top-1	58.594%	0.977%	pass
Target cosine	0.71301	0.08612	pass
Pixel F1	0.43849	0.28001	fail (< 0.50)
Ink fraction	0.15947	0.15967	controlled
Human readability	rejected	rejected	fail

Table 2: Untouched V17 evaluation with identical style and noise in each paired branch. Causal visual identity succeeds while readable topology fails.

passes every automatic development topology gate and makes many held-out forms readable, but leaves dense-form and prospective human-readability gates unresolved. The combined result establishes that a small image-only student can learn causal language signal and that a small continuous state can control recognizable generated ink; it does not establish useful autonomous language. The next proof should isolate the multiscale memory contribution while a spatial retinal field preserves dense topology. The final closed 32-cell loop must then remain readable. This order is also consistent with measured error accumulation in iterative diffusion chains [32].

13 Conclusion

The experiments show that merely replacing tokens with pixels is insufficient. A page flow can learn paper statistics, a causal writer can learn local ink, and a retina can learn character identity without acquiring language dynamics. RFLM contributes a strict test harness for ordered visual reading, direct ink flow, and autonomous rereading, but V6 and V7 reject exposure-bias correction and a monolithic language-plus-rendering writer as sufficient solutions. PVF then factors perception, causal language state, stochastic visual form, and actuation. V16’s 6.26% proposal and 3.69% state-flow top-1, positive normalized history gains, and random/unigram/last-only controls replicate the accepted visual-state proof in this project. They show that causal language signal can be learned directly from writing images by a 16.47M model on one consumer GPU. V17 then reaches 58.59% frozen visual-identity retrieval versus 0.98% under shuffled intended states with only 5.73M trainable parameters and 1.588 GiB peak memory. This breaks the categorical claim that lightweight continuous visual intent cannot control generated writing. Its 0.438 pixel F1 and

V18 development motor-plan metric	Correct intent	Shuffled intent	Gate
Global identity top-1	73.633%	0.977%	pass
Target cosine	0.84618	0.07162	pass
Pixel F1	0.65772	0.31290	pass (> 0.60)
Ink fraction	0.17505	0.17505	controlled
Human readability	simple/medium clear	unrelated	dense forms fail
Frozen images instantiated	no		sealed

Table 3: Fresh V18 development audit with style held fixed and only continuous intent shuffled. Automatic topology gates pass, but frozen promotion is withheld because the human rubric was not numerically prespecified.

unreadable pseudo-characters reject the stronger claim. V18 then uses only 2.36M trainable parameters and 0.778 GiB peak memory to reach 0.658 pixel F1 and 73.63% identity top-1 on fresh development renderings, versus 0.313 and 0.98% when intent is shuffled. This demonstrates that structured, recognizable writing can be generated as continuous image topology by a compact model on one consumer GPU without token unembedding. Dense forms still fail, the V18 frozen bank remains sealed, and the intended state is supplied rather than predicted. The seven-context V16 increase is too small to establish an architecture win, and the 13.14% bigram still rejects the causal core. The next attributable tests are paired multiscale-memory ablation and a spatial-retinal-field motor planner with a prospectively fixed readability rubric, followed only then by a closed write-reread loop. Historical etymology answers and broad instruction following remain downstream goals, not present capabilities, and no Qwen-8B parity or end-to-end efficiency claim is warranted.

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